

Pranad A. Vaishnav

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CAREER OBJECTIVE

Highly motivated and detail-oriented Computer Science student with a strong foundation in programming and software development. Passionate about leveraging technology to solve real-world problems and exploring fields such as Artificial Intelligence, Machine Learning, and Cloud Computing. Continuously developing technical skills through hands-on projects while seeking opportunities to apply and expand my knowledge in innovative and impactful environments.

EDUCATION

Bachelor of Engineering

Chandigarh University

B.E in Computer Science

Chandigarh, Punjab
Expected Graduation, May 2028

Higher secondary

State Board Maharashtra

with psychology

Pune, Maharashtra
2022 -2024 PCM

Secondary education

Air force school

CBSE board

Pune, Maharashtra
2020-2022

PROJECTS

LLM from Scratch

Technologies: Python, PyTorch, NumPy, tiktoken

Tools: Jupyter Notebook, VS Code

- ❑ Designed and implemented a **GPT-2 style decoder-only Transformer architecture from scratch**, including BPE tokenization, token and positional embeddings, self-attention, **causal masking**, multi-head attention, LayerNorm, **GELU activations**, feed-forward networks, and residual Transformer blocks.
- ❑ Built custom data pipelines for language modeling by creating sliding-window datasets and **next-token prediction workflows**, enabling efficient preparation of training sequences for **autoregressive learning**.
- ❑ Developed end-to-end training and inference pipelines using Cross-Entropy loss, AdamW optimization, validation loss monitoring, and text generation techniques such as **temperature scaling** and **top-k sampling**, gaining hands-on experience with the foundational mechanisms behind modern Large Language Models.

Vision Transformer and CNN-based Fashion Classification

Technologies: Python, TensorFlow, Keras, Vision Transformers (ViT), NumPy

Tools: Jupyter Notebook, VS Code

- ❑ Developed Vision Transformer (ViT) and CNN models for fashion image classification using custom handbag and shoe datasets.
- ❑ Implemented patch embeddings, **multi-head self-attention**, data augmentation, and **transfer learning** techniques.
- ❑ Compared Transformer and **CNN architectures** to evaluate classification performance and model generalization.

Real-Time Fashion Object Recognition

Technologies: Python, TensorFlow, Keras, ResNet50, OpenCV

Tools: Jupyter Notebook, VS Code

- ❑ Built a real-time handbag and shoe classification system using **ResNet50 transfer learning**.
- ❑ Developed image preprocessing and webcam-based inference pipelines for live object recognition.
- ❑ **Fine-tuned pretrained ImageNet** features to improve classification accuracy on custom datasets.

Seismic Data Processing

Technologies: Python, NumPy, SciPy, Matplotlib

Tools: Jupyter Notebook, VS Code

- ❑ Implemented seismic signal processing techniques including filtering, denoising, spectral analysis, and waveform visualization for geophysical data interpretation.
- ❑ Applied numerical and time-series analysis methods to enhance seismic signal quality and identify meaningful subsurface patterns.
- ❑ Built data processing pipelines for efficient analysis and visualization of large-scale seismic datasets.

Event Management Web Application

Technologies: HTML, CSS, JavaScript, Node.js, Express.js, MongoDB Atlas, Docker, Jenkins, AWS EC2

Tools: VS Code, Git, GitHub

- ❑ Developed a full-stack event management platform enabling event creation, participant registration, and user authentication through a responsive web interface.
- ❑ Designed and implemented RESTful APIs using Node.js and Express.js, integrating MongoDB Atlas for persistent storage and efficient event, participant, and user data management.
- ❑ Containerized the application using Docker and deployed it on AWS EC2, ensuring consistent and portable application environments across development and production.
- ❑ Built an automated CI/CD pipeline using Jenkins and GitHub, enabling continuous integration, Docker image builds, and automated application deployment on code updates.

Co-Curricular AND Achievements

Problem Solving & Algorithms

- ❑ Practicing Data Structures & Algorithms since first semester with focus on time complexity, optimization, and problem-solving techniques. Major platforms include leetcode, Hackerrank, Codeforce.
- Experience implementing loops, searching, sorting, graph traversal, recursion, and dynamic programming algorithms.

Certifications

- ❑ AI/ml workshop held at Chandigarh University
- ❑ Participated in Google cloud learning platform with GDG community of Chandigarh University and secured the certificate and swags

Participation in hackathons and ideathons

- ❑ Secured multiple winning and finalist positions. Collaborated in team-based problem-solving environments

Chandigarh, Punjab

TECHNICAL SKILLS

Programming Languages: Java, Python, JavaScript, Cpp, C

Web Development: HTML, CSS, Node.js, Express.js

Machine Learning: PyTorch, TensorFlow, Keras, NumPy, Pandas, Matplotlib

DevOps & Tools: Git, GitHub, Maven, Docker, Jenkins, AWS EC2

Computer Science Fundamentals: Data Structures & Algorithms (Arrays, Linked Lists, Trees, Graphs, Dynamic Programming)

IDEs: IntelliJ IDEA, VS Code, Jupyter Notebook

HOBBIES

Video editing, Debate and public speaking, Art, Reading.